WORKED SOLUTION

STEP I 2019, Question 9

Mechanics • Toppling and Friction

STEP I 2019, Question 9

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

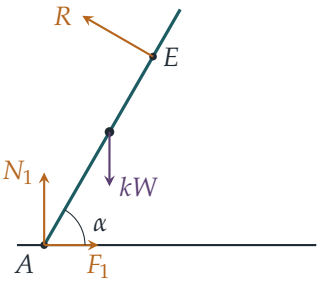
www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

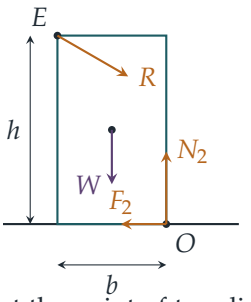
Treat the ladder and the box as two separate bodies joined by one force. Because the contact is smooth, that force is perpendicular to the ladder, so its line is fixed by α and only its size is unknown. Taking moments for the ladder about its foot gives the size; taking moments for the box about the edge it tips over gives the toppling condition. The relation $h = b\tan\alpha$ is there to make the box's moment collapse into a double-angle expression.

Solution

Model the box as a uniform rigid cuboid, the ladder as a light rod and the painter as a particle. The contact between ladder and box is smooth; both ground contacts are rough. Let E be the edge the ladder touches, A the foot of the ladder and O the bottom edge of the box furthest from the ladder (the edge about which it will tip).



ladder



box (at the point of toppling)

Part (i)

Since there is no friction at E, the force R of the box on the ladder is perpendicular to the ladder. Along the ladder, $AE = h/\sin\alpha$, because E is at height h. When the painter is at height λh they are a distance $\lambda h/\sin\alpha$ along the ladder, so their horizontal distance from A is $\lambda h\cot\alpha$.

The ladder is light, so the only moments about A come from R and the painter's weight (N_1 and F_1 act at A). Since R is perpendicular to AE, its moment arm is AE itself:

$$R\cdot\frac{h}{\sin\alpha} = kW\cdot\lambda h\,\frac{\cos\alpha}{\sin\alpha} \quad\Longrightarrow\quad R = k\lambda W\cos\alpha,$$

as required. (Strictly this holds at every stage of the climb, with λh the painter's current height.)

Part (ii)

By Newton's third law the ladder pushes on the box at E with a force R perpendicular to the ladder, directed into the box. Its components are $R\sin\alpha$ horizontally (away from the ladder, towards O) and $R\cos\alpha$ vertically downwards.

When the box is on the point of toppling about O, the ground's reaction acts entirely at O, so N_2 and F_2 have no moment about O. Take moments about O, with the clockwise sense (the way the box would tip) positive. E is a horizontal distance b from O and a height h above it, so

- the horizontal component $R\sin\alpha$, acting at height h, has moment $+Rh\sin\alpha$;
- the vertical component $R\cos\alpha$, acting a distance b to the ladder side of O, has moment $-Rb\cos\alpha$ (pushing down on that side resists tipping);
- the weight W, acting a distance $\frac{1}{2}b$ from O, has moment $-\frac{1}{2}Wb$.

The box starts to topple when the total moment reaches zero:

$$R(h\sin\alpha - b\cos\alpha) = \tfrac{1}{2}Wb.$$

Ask yourself How does $h = b\tan\alpha$ help here?

Substituting it,

$$h\sin\alpha - b\cos\alpha = b\,\frac{\sin^2\alpha - \cos^2\alpha}{\cos\alpha} = -\frac{b\cos 2\alpha}{\cos\alpha}.$$

Using part (i), $R = k\lambda W\cos\alpha$, the condition becomes

$$-k\lambda W\cos\alpha\cdot\frac{b\cos 2\alpha}{\cos\alpha} = \tfrac{1}{2}Wb \quad\Longrightarrow\quad 2k\lambda\cos 2\alpha + 1 = 0,$$

as required. Notice that this needs $\cos 2\alpha < 0$, i.e. $\alpha > 45°$, which agrees with $h > b$.

Part (iii)

Resolving for the box (whose ground reaction has components F_2 towards the ladder and N_2 upwards):

$$F_2 = R\sin\alpha, \qquad N_2 = W + R\cos\alpha.$$

The box doesn't slip, so $F_2 \le \mu N_2$, i.e. $\mu \ge \dfrac{R\sin\alpha}{W + R\cos\alpha}$.

? **Ask yourself** At which moment of the climb is this condition hardest to satisfy?

The ratio $\dfrac{R\sin\alpha}{W + R\cos\alpha} = \dfrac{\sin\alpha}{W/R + \cos\alpha}$ increases with R, and R grows as the painter climbs. So the greatest demand on friction is at the moment of toppling, and we need the inequality there. From (ii), $k\lambda = -\dfrac{1}{2\cos 2\alpha}$, so

$$R = -\frac{W\cos\alpha}{2\cos 2\alpha}.$$

Then

$$R\sin\alpha = -\frac{W\sin\alpha\cos\alpha}{2\cos 2\alpha} = -\frac{W\sin 2\alpha}{4\cos 2\alpha},$$

$$W + R\cos\alpha = W\left(1 - \frac{\cos^2\alpha}{2\cos 2\alpha}\right) = W\,\frac{4\cos 2\alpha - (1+\cos 2\alpha)}{4\cos 2\alpha} = W\,\frac{3\cos 2\alpha - 1}{4\cos 2\alpha},$$

using $2\cos^2\alpha = 1 + \cos 2\alpha$. Dividing,

$$\mu \ge \frac{-\sin 2\alpha}{3\cos 2\alpha - 1} = \frac{\sin 2\alpha}{1 - 3\cos 2\alpha},$$

as required. Both numerator and denominator are positive, since $45^\circ < \alpha < 90^\circ$ gives $\sin 2\alpha > 0$ and $\cos 2\alpha < 0$.

Checks

- Take $\alpha = 60^\circ$, so $\cos 2\alpha = -\frac{1}{2}$. Part (ii) gives $k\lambda = 1$, so $R = \frac{1}{2}W$. Then $F_2 = \frac{\sqrt{3}}{4}W$ and $N_2 = \frac{5}{4}W$, giving $\mu \ge \frac{\sqrt{3}}{5} \approx 0.346$, which matches $\sin 120^\circ / (1 - 3\cos 120^\circ) = \frac{\sqrt{3}}{2} / \frac{5}{2}$.
- As $\alpha \to 45^\circ$ (a cube-like box only just taller than wide), $\cos 2\alpha \to 0$ and (ii) needs $k\lambda \to \infty$: the push at E is almost along the line EO, so it has almost no turning effect. That matches the geometry: with $h = b\tan\alpha$, the line of R through E is close to passing through O when α is near 45°.
- The moment equation, the toppling height and the friction bound were also confirmed by computer for many random choices of α, b, k and W.

Common slips

- Taking the force at E as horizontal (as for a smooth wall). Here the contact is with an edge, so the force is perpendicular to the ladder.
- Forgetting that the vertical component of R adds to the box's normal reaction, so that $N_2 = W + R\cos\alpha$, not W.
- Taking moments about the wrong bottom edge of the box: it tips about the edge away from the ladder.

Beyond STEP

This is two-body statics: isolate each body, use Newton's third law at the contact, and pick moment centres that remove unknowns. The questions of whether a body slides or topples first, and which constraint fails first as loading increases, run through engineering mechanics and the study of rigid bodies in contact. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and friction in this spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.